

Logic PhD Exam, May 2013

Answer six questions and at least one from each section.

Section 1

1. Show that the theory of the class of infinite sets is complete and decidable, but is not finitely axiomatizable.
2. State and sketch a proof of the Łoś theorem about ultraproducts of models
3. Show that for any elementary chain $\{A_i\}_{i \in \omega}$ of structures, A_i is an elementary submodel of the union A .

Section 2

4. State and prove the Schröder–Bernstein Theorem
5. Show that Zorn's Lemma implies the Well-Ordering Principle.
6. Show that for any notion of forcing $\mathcal{P}$ and any countable set $\mathcal{D}$ of $\mathcal{P}$ -dense sets, there exists a $\mathcal{D}$ -generic $\mathcal{P}$ -filter.

Section 3

7. Prove that if $X \subseteq \omega$ has an infinite computably enumerable subset A , then it has an infinite computable subset B .
8. Sketch the proof that there are $\leq_m$ -incomparable c. e. sets.
9. Explain why the set WF of well-founded trees in ω^* and the set WO of well-orderings of ω are Π_1^1 complete and prove the Boundedness Principle, that any Σ_1^1 subset of WO is bounded below ω_1 .